

BOLD - PPI correlation analyses

Libraries and Functions

```
# Clear environment
rm(list = ls())

# Get Libraries and Functions
source("~/Dropbox (Personal)/TI_paper/ti-paper/Data/Functions.R")
```

Import data from BOLD amplitude Model 1

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/ROI/Hippocampus")
BOLD <- read.csv("BOLD_Hipp_model1.csv")

# Assign factors
BOLD$$S_ID <- factor(BOLD$$S_ID)
BOLD$ID <- factor(BOLD$ID)
BOLD$ROI <- factor(BOLD$ROI)
BOLD$hemisphere <- factor(BOLD$hemisphere)
BOLD$StimType <- factor(BOLD$StimType)
BOLD$TaskStage <- factor(BOLD$TaskStage)

# select variables/conditions of interest
BOLD_m1 <- subset(BOLD, TaskStage=="Encode" & ROI!="Hippocampus" & hemisphere=="L")
BOLD_m1 <- subset(BOLD_m1 , select = -c(hemisphere))

# change ROI variable to seed to later combine with PPI in the same dataframe
BOLD_m1 <- BOLD_m1 %>%
  rename(
    seed = ROI)
```

Import data PPI

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/PPI")
PPI <- read.csv("PPI.csv")

# Assign factors
PPI$$S_ID <- factor(PPI$$S_ID)
PPI$ID <- factor(PPI$ID)
PPI$StimType <- factor(PPI$StimType)
PPI$TaskStage <- factor(PPI$TaskStage)
PPI$network <- factor(PPI$network)
```

```

PPI$seed <- factor(PPI$seed)
PPI$target <- factor(PPI$target)

PPI <- subset(PPI, TaskStage=="Encode" & network=="AT")

# Average all targets from AT
PPI_AT <- ddpoly(PPI, c("S_ID","ID","StimType","seed","TaskStage"), summarise,
  conn = mean(mcon))

```

Perform Correlations

```

# Combine dataframes
all <- merge(BOLD_m1,PPI_AT,by=c("S_ID","ID","StimType", "TaskStage","seed"))

library(data.table)
# Correlations - are the changes in BOLD amplitude and connectivity correlated?
corr1 <- ddpoly(all, .(StimType, seed), corrgroup, "BOLD", "conn")
pander(corr1)

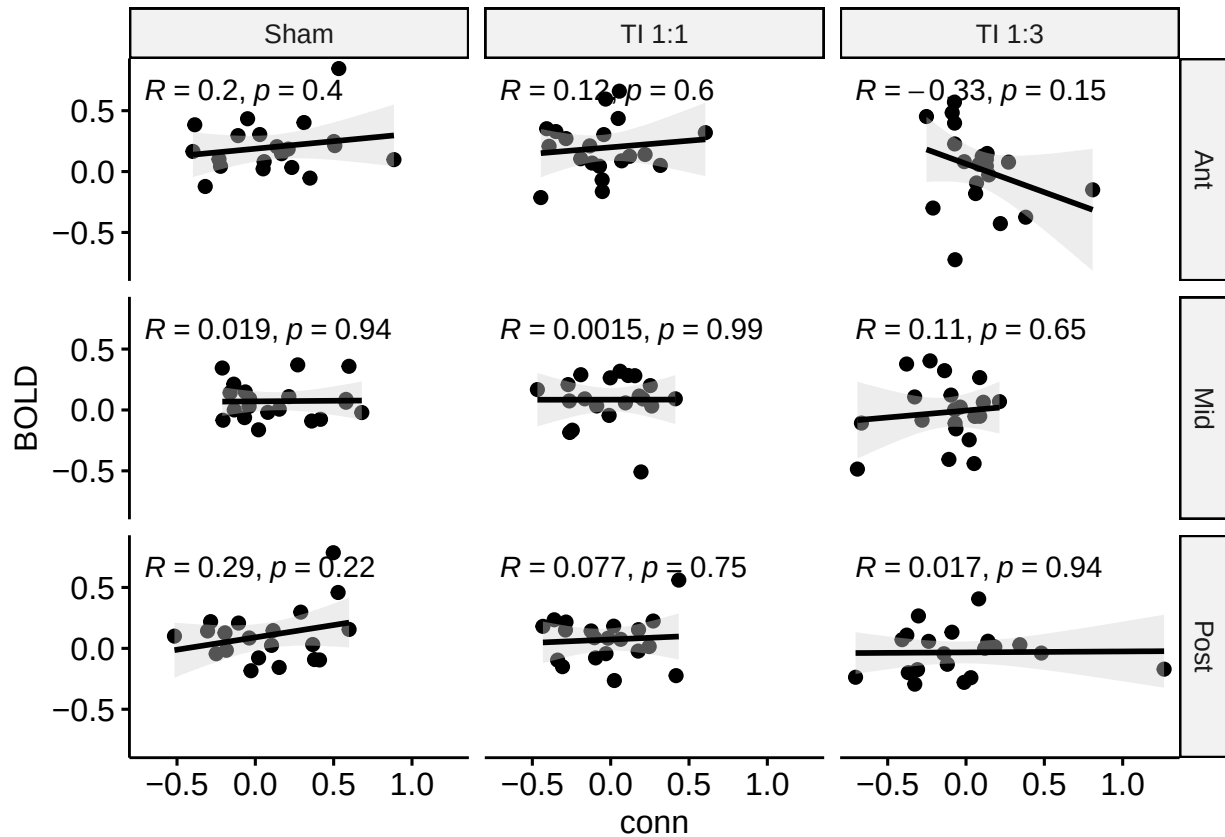
```

StimType	seed	COR	p.value
Sham	Ant	0.198	0.4026
Sham	Mid	0.01886	0.9371
Sham	Post	0.2861	0.2214
TI 1:1	Ant	0.1245	0.6009
TI 1:1	Mid	0.001537	0.9949
TI 1:1	Post	0.07678	0.7477
TI 1:3	Ant	-0.3304	0.1548
TI 1:3	Mid	0.1091	0.6472
TI 1:3	Post	0.01736	0.9421

```

# Plot
sp <- ggscatter(all, x = "conn", y = "BOLD",
  add = "reg.line", # Add regressin line
  add.params = list(color = "black", fill = "lightgray"), # Customize reg. line
  conf.int = TRUE # Add confidence interval
) + stat_cor(method = "pearson") + facet_grid(seed~StimType)
sp

```



Import data from BOLD amplitude Model 2

```
setwd("~/Dropbox (Personal)/TI_paper/ti-paper/Data/MRI/FaceNameTask/ROI/Hippocampus")
BOLD <- read.csv("BOLD_Hipp_model2.csv")

# Assign factors
BOLD$$S_ID <- factor(BOLD$$S_ID)
BOLD$ID <- factor(BOLD$ID)
BOLD$ROI <- factor(BOLD$ROI)
BOLD$hemisphere <- factor(BOLD$hemisphere)
BOLD$StimType <- factor(BOLD$StimType)
BOLD$TaskStage <- factor(BOLD$TaskStage)
BOLD$Response <- factor(BOLD$Response)

# select variables/conditions of interest
BOLD_m2 <- subset(BOLD, TaskStage=="Encode" & ROI!="Hippocampus" & hemisphere=="L" & Response=="C>I")
BOLD_m2 <- subset(BOLD_m2, select = -c(hemisphere, Response))

# change ROI variable to seed to later combine with PPI in the same dataframe
BOLD_m2 <- BOLD_m2 %>%
  rename(
    seed = ROI)
```

Perform Correlations

```
# Combine dataframes
all2 <- merge(BOLD_m2,PPI_AT,by=c("S_ID","ID","StimType", "TaskStage","seed"))

# Correlations - are the changes in BOLD amplitude and connectivity correlated?
corr2 <- ddpby(all2, .(StimType, seed), corrgroup, "BOLD", "conn")
pander(corr2)
```

StimType	seed	COR	p.value
Sham	Ant	0.3038	0.1928
Sham	Mid	0.5036	0.0236
Sham	Post	0.01802	0.9399
TI 1:1	Ant	0.008081	0.973
TI 1:1	Mid	0.2636	0.2614
TI 1:1	Post	0.2488	0.2901
TI 1:3	Ant	0.08913	0.7086
TI 1:3	Mid	-0.2426	0.3028
TI 1:3	Post	0.2411	0.3058

```
# perform correction for multiple comparisons
ost_FDR(corr2)
```

```
##   StimType seed      COR   p.value    FDR  sig
## 1   Sham  Ant  0.303808977 0.19283788 0.4587073 <NA>
## 2   Sham  Mid  0.503570465 0.02359875 0.2123887  *
## 3   Sham Post  0.018018625 0.93989740 0.9730269 <NA>
## 4  TI 1:1  Ant  0.008080877 0.97302693 0.9730269 <NA>
## 5  TI 1:1  Mid  0.263638211 0.26138501 0.4587073 <NA>
## 6  TI 1:1 Post  0.248810641 0.29013844 0.4587073 <NA>
## 7  TI 1:3  Ant  0.089129002 0.70864487 0.9111148 <NA>
## 8  TI 1:3  Mid -0.242555383 0.30282906 0.4587073 <NA>
## 9  TI 1:3 Post  0.241111843 0.30580483 0.4587073 <NA>
```

```
# Plot
sp2 <- ggscatter(all2, x = "conn", y = "BOLD",
  add = "reg.line", # Add regressin line
  add.params = list(color = "black", fill = "lightgray"), # Customize reg. line
  conf.int = TRUE # Add confidence interval
) + stat_cor(method = "pearson") + facet_grid(seed~StimType)

sp2
```

